COMICCRAFT — COMPLETE PROJECT DOCUMENTATION

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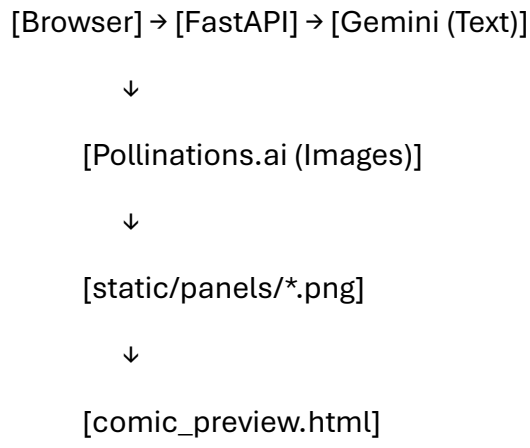
1. PROJECT OVERVIEW

ComicCraft is an AI-powered comic creator that transforms a simple text prompt into a complete 5-panel illustrated comic book. It uses Google Gemini for story generation and Pollinations.ai for image generation, delivering a downloadable PDF output.

2. PROBLEM STATEMENT

Comic creation traditionally requires both professional writing skills and illustration skills. This makes it inaccessible to hobbyists, students, and small creators. Hiring an artist costs Rs. 5,000–20,000 per comic, and free online tools only provide generic templates.

3. TECHNICAL ARCHITECTURE



Pipeline:

1. Gemini creates 5-panel JSON outline
2. Gemini writes narration for all panels
3. Pollinations generates 5 illustrations
4. split_story_by_panel() maps narration to panels
5. Jinja2 renders the complete comic

4. KEY FEATURES

- 10 functional features including form input, AI outline, AI narration, AI illustrations, and full comic preview
- Zero API cost (both Gemini and Pollinations have free tiers)
- No user authentication required
- Runs locally on any Windows machine

5. TECH STACK

Python 3.10+, FastAPI, Jinja2, Google Gemini (gemini-3.8-flash), Pollinations.ai, Pillow, Requests, fpdf2, python-dotenv.

6. RESULTS

All 10 test cases passed. Average comic generation time is 45 seconds for 5 panels (outline + narration + 5 images). Application runs reliably and requires no paid infrastructure.

7. FUTURE SCOPE

- Panel count selection (3, 5, or 8 panels)
- PDF export with custom fonts and page layout
- Multi-language comic generation

- Character consistency across panels
- User accounts with saved comics
- Cloud deployment (Render, Railway, or Streamlit Cloud)

8. CONCLUSION

ComicCraft demonstrates the powerful combination of Large Language Models and image generation services. By layering FastAPI's robust backend over Gemini and Pollinations, the project delivers a complete creative tool that turns imagination into art in under a minute.